Vector addition

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add displacement vectors and use a model to explain where an object ends up.

SUCCESS CRITERIA - I can...

1. Draw and label arrows tip to tail; keep their length and direction.
2. Draw the resultant from the original start to the final endpoint.
3. State net east-west and north-south motion with units and direction.

Core: teacher selects from 1-10. **Stretch:** 11-12 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “vector” and “component” to explain your model or result. Sentence starters are optional.

vector: a quantity with size and direction

component: the part of a vector along one chosen axis

resultant: one vector equal to the sum of other vectors

01 Bring back an earlier idea

A cart moves from 0 m to 8 m east in 4 s at a steady pace.
Find its speed.
Draw a velocity arrow showing its direction.
Explain what the arrow tells you that speed alone does not.

Word help: velocity = rate of position change, with direction

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Read and explain a worked model

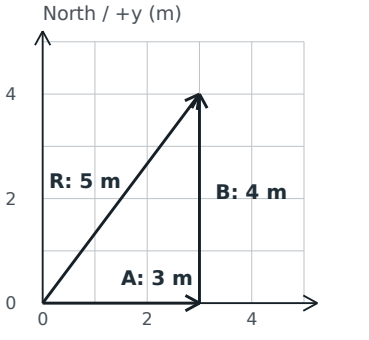
Reference: $R_x = A_x + B_x$; $R_y = A_y + B_y$.

Worked reasoning: A: 3 m east. B: 4 m north. R_x=3 m; R_y=4 m. Path=3+4=7 m. The supplied straight-line length is 5 m, so displacement is 5 m northeast.

Your turn: A walker goes 3 m east, then 4 m north.
Circle the start and endpoint.
Explain why 7 m is distance, not displacement magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: My claim is … . The diagram/data show …, so … .

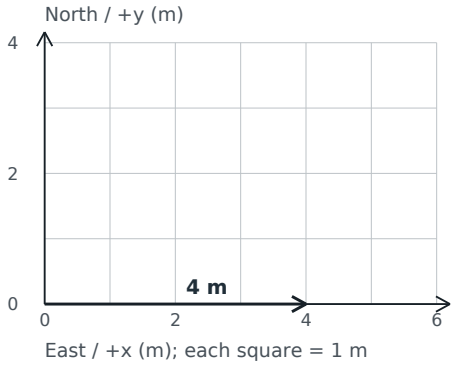


Marzano check - circle: ① ② ③

03 Finish the model

Walk 4 m east, then 3 m north. The first arrow is drawn. Add the second arrow from its tip and a dashed resultant from the original start.
Give net eastward and northward motion. The straight-line length is 5 m: state displacement and direction.
Find distance traveled.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 A trip that doubles back

Walk 6 m east, then 4 m west.
Draw and label the route and resultant.
Give distance and displacement.
Explain which length follows the path.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

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Continue down each column.

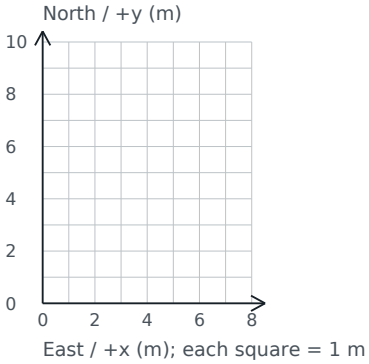
Sentence starters are optional. You may plan in any language.

05 Use a new scale

Walk 6 m east, then 8 m north.
Draw both arrows and the resultant.
Give distance and displacement. The straight-line length is supplied: 10 m.
Show the start, turn and endpoint; label units.
Explain how the drawing shows displacement magnitude is less than distance.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

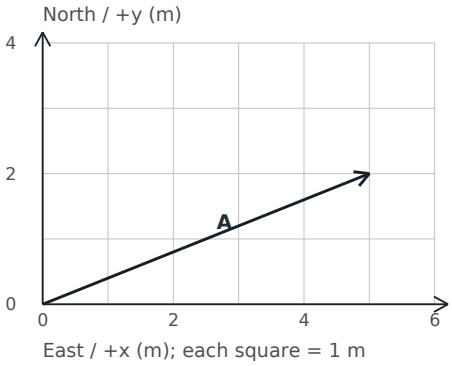


Marzano check - circle: ① ② ③

06 Add two diagonal arrows

A: 5 m east and 2 m north. B: 3 m west and 1 m north. A is drawn. Add B and R.
State each net component and the general compass direction.

Sentence starter: The … in my model represents … . Its label shows … .

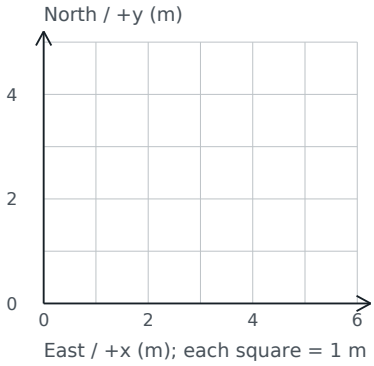


Marzano check - circle: ① ② ③

07 Predict, draw, check

Walk 3 m east, 4 m north, then 3 m west.
Predict what cancels.
Draw the route and R.
Give distance and displacement.
Check your prediction against the model.

Sentence starter: I predict … . My model supports this because … .



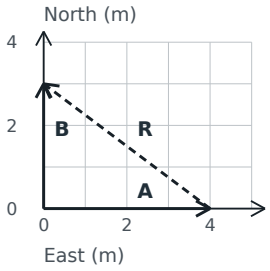
Marzano check - circle: ① ② ③

08 Find and repair the mistake

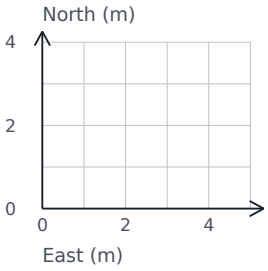
Sam adds 4 m east and 3 m north. His dashed R is wrong.
Explain why and draw the correct route and resultant on the empty grid.

Sentence starter: The step I would change is … . The evidence is … .

Sam: incorrect



Your corrected model



Marzano check - circle: ① ② ③

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Continue down each column.

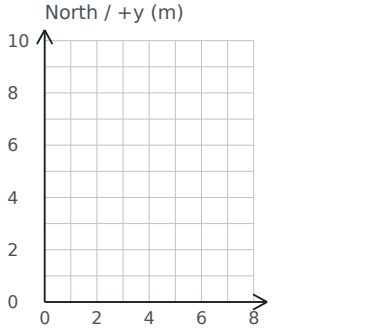
Sentence starters are optional. You may plan in any language.

09 One trip, three questions

A student walks 6 m east in 6 s, then 8 m north in 8 s. The direct start-to-finish length is 10 m. For each answer, decide whether to use path length or the start-to-finish arrow, then solve:
(a) Distance walked? (b) Displacement?
(c) Average speed for the whole trip?
Draw and label the route and resultant.
Show where total time enters the solution; include units in all answers.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.

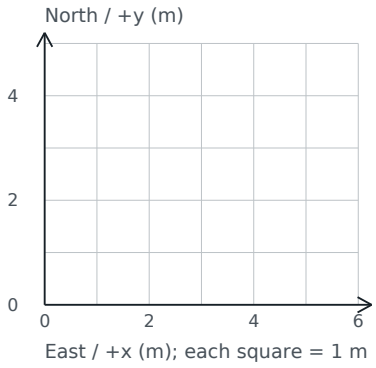


Marzano check - circle: 1 2 3

10 Exit check - work independently

A goes 4 m east and 1 m north. B goes 1 m west and 3 m north.
Draw tip to tail and add R.
Give both net components and a general compass direction.
Explain one check of your model. After feedback: star a step that supports your rating; explain any correction. Ask for help if drawing and calculation disagree.

Sentence starter: My claim is … . The diagram/data show …, so … .



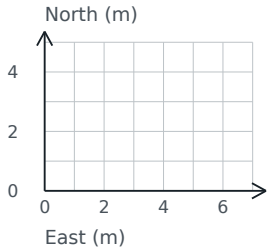
Marzano check - circle: 1 2 3

11 STRETCH: Design two routes

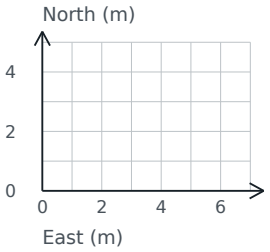
Finish 4 m east and 3 m north of your start.
Design one route of length 7 m and another of length 11 m.
Use only north, south, east or west legs.
Label all lengths and draw each resultant.
Explain how distance can differ while displacement stays the same; show a length check for each route.

Sentence starter: Both … . They differ in … because … .

Route 1: 7 m total



Route 2: 11 m total



Marzano check - circle: 1 2 3

12 STRETCH: Explore the limits

Point two 5 m displacement arrows in any directions.
Draw the smallest and largest possible resultants, then one between those extremes.
Estimate the intermediate resultant and explain why your limits make sense. Keep one scale in each sketch.
Label original start, final endpoint and resultant. A ruler is allowed; no angle calculation is required.

Word help: estimate = find an approximate value

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3